

St. Mary's Higher Secondary School

Subject – Physics

Total time – 1½ hrs.

FM- 40

Section A - MCQ (1 Mark Each)

Q1. As compared to ^{12}C atom, ^{14}C atom has:

- (a) Two extra protons and two extra electrons
- (b) Two extra protons, but no extra electrons
- (c) Two extra neutrons and no extra electrons
- (d) Two extra neutrons and two extra electrons

Q2. The ratio of the mass of a nucleus to the mass of an atom is nearly:

- (a) 1
- (b) 10
- (c) 10^3
- (d) 10^{10}

Q3. According to Bohr's model, the angular momentum of an electron in the third orbit of a hydrogen atom is:

- (a) $h/2\pi$
- (b) $2h/2\pi$
- (c) $3h/2\pi$
- (d) $4h/2\pi$

Q4. The first line of the Balmer series ($H\alpha$) is obtained when an electron jumps from:

- (a) 2nd orbit to 1st orbit
- (b) 2nd orbit to 3rd orbit
- (c) 3rd orbit to 2nd orbit
- (d) 3rd orbit to 1st orbit

Q5. An n-type semiconductor is one which has:

- (a) Holes as majority carriers
- (b) Free electrons as majority carriers
- (c) Holes and free electrons equal in number
- (d) Trivalent element as an impurity

Section B - Very Short Answer (1 Mark Each)

Q6. State the physical condition under which Ohm's law is not valid for a conductor.

Q7. State Bohr's postulate regarding the angular momentum of an electron.

Q8. Name the series of spectral lines in the hydrogen spectrum that lies in the ultraviolet region.

Q9. What is meant by mass defect?

Q10. Why are nuclear fusion reactions also called thermonuclear reactions?

Section C - Short Answer (2 Marks Each)

Q11. When a test charge of 2 nC is placed at the origin, it experiences a force of 8×10^{-8} N in the positive y-direction. Find the electric field at the origin and the force on a charge of -4 nC placed at the origin.

Q12. The intensity of electric field at a perpendicular distance of 0.5 m from an infinitely long line charge is $3.6 \times 10^3 \text{ Vm}^{-1}$. Calculate the linear charge density.

Q13. You are provided with identical capacitors each of capacitance 100 μF . How will you connect the minimum number of them to obtain a capacitance of 75 μF ? Draw a diagram.

Q14. Two nuclei have mass numbers in the ratio 1:2. Find the ratio of their nuclear densities.

Q15. Explain briefly why Rutherford's atomic model could not account for the stability of the atom.

Section D - Short Answer (3 Marks Each)

Q16. Two small, identical metallic spheres A and B carrying charges repel each other with a force F when placed at a distance r. After touching identical uncharged spheres C and D respectively and removing them, B is brought to r/2 from A. Find the new force of repulsion.

Q17. An α -particle is projected towards a thin gold foil from a large distance. Derive an expression for the distance of closest approach.

Q18. Write a balanced nuclear equation representing fission of uranium. State the functions of (a) cadmium rods (b) graphite rods in a nuclear reactor.

Q19. Using Gauss's theorem, obtain an expression for electric field intensity outside a charged spherical shell.

OR

Three point charges 25 μC , 50 μC and 100 μC are placed at the vertices of an equilateral triangle of side 7.5 m. Calculate the total electrostatic potential energy.

Q20. With reference to semiconductors: change in resistance with temperature, majority carriers in n-type semiconductor, and define doping.

OR

Calculate the mass defect and binding energy of ^{20}Ne nucleus given the required masses.

Section E - Long Answer (5 Marks Each)

Q21. Obtain an expression for intensity of electric field at the end-on (axial) position of an electric dipole.

OR

State the postulates of Bohr's model of hydrogen atom. Hence derive the expression for the radius of the nth orbit of an electron.